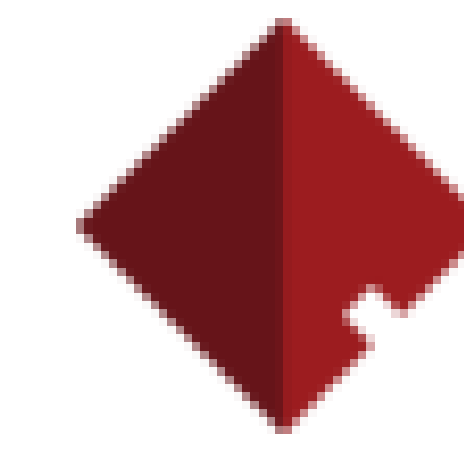




Robustness Testing for Self-Driving Cars



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Introduction

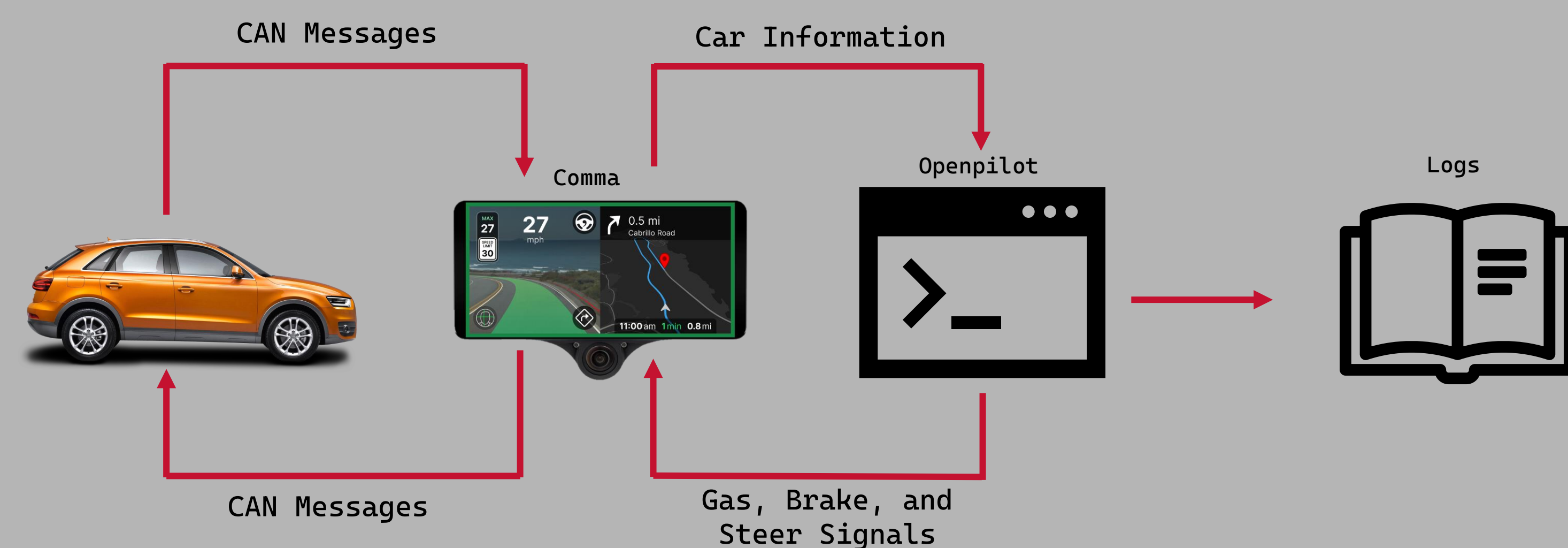
What is Robustness Testing?

- A type of software testing that tests a system under rare, off-nominal, and extreme conditions.



What is Openpilot?

- An opensource software that runs on a device called a Comma. The Comma device utilizes a car's CAN bus to turn it into a level 2 driving assistant. This means the device can control steering, acceleration, and braking, but the driver must still pay attention to the road.



Why does Openpilot need Robustness Testing?

- Openpilot can be installed into 250+ cars, and the software can be contributed to by anyone. Such an accessible piece of technology needs more safety guarantees than it currently provides. In utilizing robustness testing for Openpilot, it can be tested in edge cases that are not typically considered.

Related Works

Previous Works

- Various researchers have tested Openpilot via fault injection into the vision system [1][2][3][5], seeing how Openpilot reacts to adversarial driving maneuvers[4], and applying falsifiers to the neural network they use to predict routes[6].

Literature Gaps

- Many works have relatively binary methods of measuring failure, saying it's a failure if the car crashes and success if not. As well as not including factors about the surrounding environment in their simulations. This does not comply with the more comprehensive testing recommended by safety professionals.

Current Work and End Goals

Log Reading

- Openpilot logs the various information about the car's state, from things like the velocity and acceleration of the vehicle, to the driver's attention and the leading vehicle.
- All the information logged by Openpilot can be extremely useful for understanding what the car sees as well as

Invariant Monitors

- From the information parsed from the logs, invariant monitors based on safety standards can be applied to ensure that the car remains safe operations.



Introducing Perturbations

- Using the software MoBSTA (Mutation Based Safety Testing for Autonomy), perturbations can be applied to the logs, and then the logs can be replayed on Openpilot to see if it reacts properly to



Car
Route



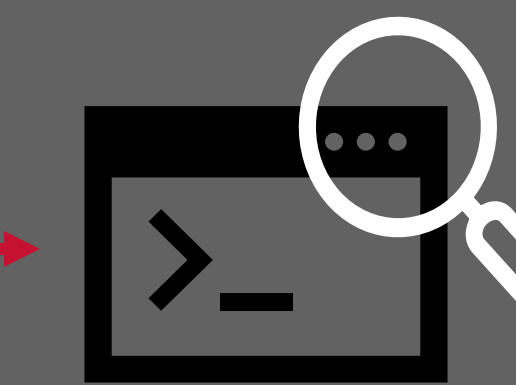
Logs



Perturbations
Applied



Logs with
irregular
data



Logs replayed through
Openpilot and
monitored



30-mph
70 mph

Distance
15-ft
2 ft



30-mph
12 mph

- Will Openpilot still regard this as safe?

Conclusion

Overall, the goal of this project is to apply robustness testing to the software Openpilot as a means of checking it against current Autonomous Vehicle safety standards. Previous works have tested Openpilot through various software testing methods, but many had a rather binary way of accessing if the software behaved safely, and some were not using metrics informed by current safety standards. In utilizing robustness testing for Openpilot, the software can be testing in a comprehensive manner.

References

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